INF632 Research Project

Build your Own Wearable / Analysis of Commodity Wearable Data

1 Literature Review

Assigned: February 10.

Due: February 24 at 11:59pm. Submissions turned in after this time and still within two weeks of this deadline will be automatically docked 50% of the possible points.

Submission: Share a direct link to your repo / folder on canvas.nau.edu by the deadline.

Points: EE499 students, this is worth 5% of your final grade. EE599 students, this is worth 10% of your final grade.

Background:

Science is fundamentally rooted in the ongoing pursuit of knowledge, which is reflected in the existing literature. A thorough understanding of this body of work is essential for identifying gaps and formulating the pertinent questions that should be explored next. To effectively chart a course for further research, one must engage deeply with what has already been accomplished in the field. Failing to do so may lead to redundant efforts, essentially reiterating the findings of previous studies rather than contributing new insights. By critically analyzing existing literature, researchers can ensure that their work builds upon established knowledge, drives innovation, and addresses unanswered questions in the scientific community.

Assignment:

This is the first of five assignments related to your class project. In this assignment, you need to identify papers related to your desired project. These should provide background and motivation for what you want to do within your project. Up to two can directly reflect what you want to do (for a class project, it is okay to repeat the work of others).
At least three citations must be discussed in detail; others can be "lightly" reviewed. Write a summary of the "current state of the art" related to your research project. See the papers we have discussed and the "extras" to get a good feeling for what this section should look like. Make sure that you are clearly motivating the direction for your project from the literature you review.

Expectations and Grading Rubric:

I expect your literature review to be complete, on the order of three to twelve paragraphs. You submission must use the IEEE JBHI Template (Grad must use LaTeX, Undergrad may use the LaTeXor Word template).
On the topic of ethics, be sure that you have properly cited everything and have not plagerized anyone's work in your paper. You may use a generative AI to help you revise paragraphs or to help with detailed guidance, but do NOT use it to write your paper. You're here to learn - please go through the process of learning!
Your submission will be graded as follows:

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INF 632
Research Project – Part 1

Given that obesity is a growing issue around the world, there is a driving need to discover a solution to reduce obesity rates. One solution that has been proven to help obese individuals lose and keep weight off is to self-monitor the intake of food during meals and consumption. Although there have been a number of proposed ideas to track intake (including food diaries, questionnaires, and consumption recalls), one popular area of research involves the use of automated wearable devices to detect periods of food intake. In particular, wrist-worn devices are of interest due to their unobtrusiveness and user friendliness. The ability to accurately and cost effectively detect food intake during a period of consumption has the potential reduce the obesity rates and improve health conditions all over the world.

Although this is a growing area of interest, there is currently no substantial "gold-standard" for measuring the intake of food via wrist-worn devices. This being said, there have been a number of researchers who have developed ideas and tested devices to do this, particularly related to bite detection and bite count. Dong et al. [1] developed a single wrist-worn sensor that emphasized focus on the rolling motion of the user's wrist. Using the InertiaCube3 sensor produced by InterSense Corporation and a self-developed algorithm, they were accurately able to detect bites of food at a sensitivity of 91%. Similarly, Dong et al. [2] proposed a method using a watch-like device that could be turned on and off before and after eating. Again, the focus was to track wrist orientation and wrist roll to automatically detect bites of food. Given the costliness and bulkiness of the InertiaCube3 sensor, this study used the STMicroelectronics LPR530a1 to directly measure radial velocity. This device obtained similar results as the previous experiment. Mendi et al. [3] had also developed a similar device, though specific parts used were not detailed, which provided live feedback to the user. The acceleration data was sent to a mobile phone via Bluetooth and processed there to provide messages to the user. It should also be noted that studies on wrist-worn devices have been done in regards to fluid intake as detailed in Amft et al. [4].

For this research project, I would like to research and develop a wrist-worn device to detect food intake and particularly bites of food. I would use a device with a micro-electromechanical gyroscope to track wrist roll and orientation as discussed in Dong et al. [2]. I would then develop an algorithm to determine when a user is taking a bite of food. On top of this, if time (and knowledge) permits, I would like to find a way to provide live feedback to the user as discussed in Mendi et al. [3].

Citations

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Reliable Knee Flexion Measurements Using Readily Available Components

[redacted]

Lower limb kinematic analysis is a critical component in many fields and disciplines including health care, sports medicine, and biomimetic design. Diseases that affect mobility can be measured in abnormalities in the gait cycle [1] . Sports teams can compare athlete motions against expert motions to minimize joint impact and maximize efficiency [2]. Rehabilitative robotics can use measured gait data to develop models to mimic human motion [3]. Kinematic data is highly valuable but expensive to obtain.

The gold standard for motion capture is a marker-tracking optical system like the VICON camera system. VICON employs multiple cameras tracking reflective markers placed at strategic locations on a subject. Coupled with special software the VICON can give highly detailed information about each tracked limb. The VICON is extremely powerful but stationary, limiting data capture to controlled environments over a specific time frame. While sufficient for many applications there are many that need more diverse data. For example, walking disorders may have intermittent symptoms that a doctor may never observe. Furthermore, most movement does not occur in a lab and may involve motions that are unanticipated. It is therefore advantageous to develop a system that can collect kinematic data in more situations over a longer period.

Research into wearable systems that can supplement VICON data or replace it all together has shown significant promise. These systems currently use either one or a combination of the following measurements: Ground Reaction Force (GRF) through pressure transducers [4], linear and rotational accelerations via Inertial Measurement Units (IMU) [5], skin stretch through cable-potentiometer systems [6], and bend angle using an electrogoniometer [7] [8].

Tognetti A. et. al. developed a system that has demonstrated a high degree of accuracy while remaining lightweight and low-cost [7]. Though limited to only measuring knee flexion angle similar systems may be developed to include other joints. Tognetti's device used a combination of a textile goniometer and two IMUs to accurately measure knee flexion angle. The goniometer measured knee angle and the IMUs were used to actively correct the calibration of the goniometer, resulting in a system nearly as accurate as the optical reference system. A major shortcoming of this device is that it remains tethered to a stationary computing system.

Wang P. et al. developed a simple device to estimate knee flexion angle using inexpensive flex sensors [8]. They estimate that their device is at least an order of magnitude less expensive than commercial options. The device is demonstrated to be provide measurements related to angle of flexion or extension but no validation of the measurements is made. Additionally, the device is taped onto the subject, making it less than ideal to don and doff.

It is proposed that a simple, lightweight, self-contained system be developed to measure knee flexion angle. This device will have its own power supply and microcomputer for preprocessing data. Three sensors that measure flexion angle through a change in resistance will be the only sensors used. All measurements will be recorded on a microSD card for later analysis. The anticipated product will be able to run for 12 hours, record measurements at a minimum of 30hz, and be able to tolerate $\pm$ 3° in sensor orientation with no calibration required for every series of tests. It is the intent to develop a system that is easy to use and provide useful and reliable information.

References

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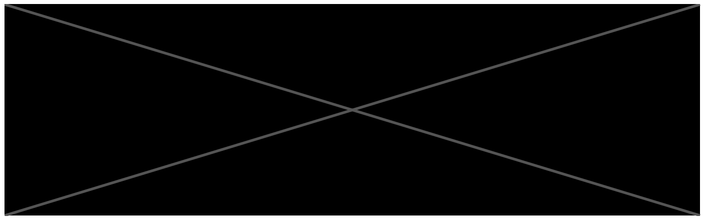
Arduino-Based, Multi-Sensor Detection and Differentiation of Static Activity



I. Introduction

Technological advancements have lead to the development of lightweight, wearable sensors that are capable of detecting the intricacies of human motion. By remotely monitoring and analyzing a person's physical state, it is possible to identify underlying behavioral patterns, create models of physical activity, uncover neurological disorders, or provide tools for lifestyle intervention. Although some physical activity monitors are commercially available, the majority of solutions are either prohibitively expensive, markedly inaccurate, or provide restrictive licenses that limit the extensibility of the platform. Additionally, these platforms largely focus on dynamic physical states such as running, walking, cycling, etc., and they pay little attention to static states such as sitting, standing, and lying down [4]. While an analysis of dynamic physical states is certainly useful in the context of physical activity (calories burned, distance walked, steps taken, etc.), an analysis of a user's static physical state may yield valuable information regarding behavioral or psychological phenomena. As such, it is important develop a robust system that is capable of classifying and analyzing not only functional movement, but also static physical states.

A. Related Work

Although commercial monitors are largely geared toward the detection of functional activities, posture detection has been performed using a variety hardware and software platforms. In [2] and [5], the authors use a single, triaxial accelerometer to determine a subject's posture (sitting, standing, lying down) as well as detecting certain motions (picking up objects, moving while lying down, etc.). Both studies result in similar success levels, reporting accuracies of 96.4% and up to 98.2%, respectively. Similarly, [4] uses a single device to monitor subjects, but differs from the first two due to the use of the "gold standard" Actigraph GT3X+ and ActivPal monitors. Rather than placing the monitors at the hip of the subjects (as is common practice with these monitors), both monitors were placed approximately two inches apart midway between the knee and hip. This study was conducted to test the ability of these monitors to detect the difference in sitting, standing, walking, standing while writing on a whiteboard, and upright cycling. As well as comparing data from the monitors to laboratory observations, the authors compared the agreement of the two monitors. The authors found high levels of agreement for many activity states, but found that some movements (writing a blackboard, upright cycling) were often misclassified as stepping.

In addition to single-sensor systems, the research community has explored the use of multi-sensor platforms to analyze posture. In [1], the authors use a series of four triaxial accelerometers and a LilyPad Arduino microcontroller to analyze a subject's spine angle. One accelerometer is placed at each of the cervical, thoracic, lumbar, and sagittal portions of a skin-tight undergarment. These accelerometers calculate the spine angle specific to their region and communicate with a mobile device via a bluetooth modem. The application compares the measurements from the accelerometers to nominal values to determine if the user's posture is indicative of cervical lordosis, thoracic kyphosis, or lumbar lordosis. Conclusions are presented to the end user through the use of a mobile application. The methods involved in the four related works are consistent with the findings of [3], which indicate that the use of multiple sensors, though consuming more power, typically result in less computationally intensive analyses.

B. Contributions

Similar to previous works, this paper presents a method for the analysis of static states. However, this work differs from others due to the special attention paid to the differentiation of active and passive forms of static activities (i.e. periods of work vs periods of relaxation). We hypothesize that different forms of static activities are accompanied by different postures, e.g. a user's posture while working at a desk is differentiable from sitting on a couch watching television. The subject's posture will be monitored through the use of an Arduino Mini Pro microcomputer and two, 3-axis accelerometer/gyroscopes placed on the upper arm and upper leg. Since a nearly infinite set of activities can be performed in each static state (sitting, standing, lying down), we consider only "active" and "passive" versions of each state rather than specifying particular actions. In future work, we plan to explore correlations between physical signs of stress and static behavioral characteristics (e.g. increased heart rate as it relates to duration/ proportion of active and passive activity states).

References

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INF 632 – Research Project - Part 1

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Predicting physical stress in different types of exercise using a wearable

Introduction

Nowadays, wearable technologies are taking importance when assessing different types of exercise in order to estimate how the amount of exercise affects a subject's health by analyzing different measurements.

Cardiovascular response to physical stress was studied through heart rate and blood pressure [1] and the results showed that in a range of intensities, the values regulation can be represented by a simple RLC model that can be useful to estimate the cardiovascular condition while exercising to prevent health issues. In this other study [2], Heart Rate Monitors (HRM) were evaluated for a large number of users classified in four groups: enthusiastic users, active users, fading users and infrequent and given-up users. The conclusions showed that two groups, the fading and infrequent users had a decreasing trend towards the usage motivation while the other two groups, the active and enthusiastic users kept the motivation all time. This decrease of usage motivation in two groups out of four is an issue as we want all the users of our wearable technologies to keep the initial motivation to use the device every day in order to assess the results.

As well as heart rate, the study of the response of the skin temperature to physical exercise can provide an improvement to detect the physical stress in different types of exercise, intensity and duration. In this review [3], a lot of studies of the skin temperature after different types of exercise, with different population and measurements were evaluated. The overall conclusion displayed two different situations for active and non-active muscles and high and low intensity activities. For the active muscles, the skin temperature increases during high intensity exercises, decreases right after it and increases again in the following days. When analyzing low intensity activities, the results showed that skin temperature decreases during exercise, returns to normal levels right after it and slightly increases the temperature in the following days. For the non-active muscles, we see no difference between different types of intensity. In this case, the skin temperature is decreasing during the activity, returning the normal levels after it and rising to whole body levels in the days after the exercise.

Other studies have inspected the exercise stress using different methods as can be using the Photoplethysmogram (PPG). In this clinical trial [4], the low frequency spindle waves from the PPG were detected in all subjects under cardiovascular stress in the forehead and ear located device. The origin of the waves is unknown, but it can be used to automatically detect exercise induced physical stress. Another different method would be to analyze the exercise stress measuring the stroke volume (SVV) and systolic blood pressure (SBPV). In this paper [5], the spectral distribution of the power of these values is concentrated in two bands, low frequency and high frequency. The results indicate that the SV components for LF, HF and LF/HF increased during exercise. However, for SBPV, no consistent change patterns were found. So, to sum up, coherence between the variables changed differently during the exercise test by the fact that stress test not only excites the sympathetic nerve system. Therefore, HF components may reflect the influence of some common impact factors while LF components mainly reflect the different mechanism of the cardiovascular functions.

From the related literature, in this study we will design a wearable device that will be able to predict physical stress in different types of exercise and intensity. We will use an accelerometer, a pulse sensor and a body temperature sensor to measure the stress and communicate with the user in order to prevent health problems while exercising.

REFERENCES

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A Review of Wearable Devices to Detect Alcohol Consumption

[redacted]

Current wearable alcohol consumption monitoring systems use ethanol sensors [1], [2], alcohol gas sensors [3], and commodity hardware [4], to detect the wearers intoxication. Three of these devices use a sensors placed on the skin [1]–[3] and one of them uses the built-in sensors of smart phones [4]. Commercially available devices also exist to monitor alcohol consumption in real-time [1][5][6]. However, there are limitations with these devices, such as subject specific calibration and detection methods, cost, lack of intervention functionality and lack of social based accountability. It should be noted that a commercial device is currently in the works that does incorporate social accountability [7], however it relies on user feedback and to our knowledge does not incorporate machine learning techniques.

In 2013, the Centers of Disease Control and Prevention (CDC) found that on average a person was killed every 51 minutes do to an intoxicated driver [8]. Also, in a study conducted in 2006 found that there was high correlation between sexual assault and alcohol consumption [9]. Preventing and intervening during these types of behaviors can have great mental and physical health benefits. Imagine if that college student was not sexually assaulted and was spared the mental trauma because her friends were notified of her intoxicated state and were able to intervene. Imagine if that innocent bystander wasn't killed by an intoxicated driver because the driver was able to assess his/her intoxication before getting behind the wheel. Obviously, the blame falls on the individual assaulter and driver, though having technologies that can prevent these actions is highly desirable.

One of the most common approaches to detecting consumption is to use transdermal ethanol sensors. A 1993 study describes the testing and verification of a Giner Inc. wearable transdermal ethanol sensor [1]. Experiments were conducted with "intoxicated" subjects (e.g. subjects who had consumed known amounts of ethanol) and with "sober" subjects (e.g. subjects who had not consumed ethanol). A control measurement was also gathered using a breathalyzer. Measureable amounts of ethanol vapor are known to perpetrate through the sweat glands and in this study was shown to closely follow the breath blood alcohol concentration curve. The consumption curve of the ethanol sensor compared to the control was found to be right-shifted by up to 120 min. This result "suggest the existence of a distinct pharmacokinetic compartment for cutaneous ethanol" [1], meaning there is a mechanism within the body that determines how the ethanol reaches the skin. This result is similar to the delay in alcohol levels found in urine. This work showed that it is possible to measure alcohol consumption using vapor sensors placed at the skin and the study is widely referenced in the field.

A less common approach is to use commodity hardware such as an Arduino and gas sensor. In 2017 Blackburn Cnaptic did exactly that by using an alcohol vapor sensor, a 3-axis accelerometer, and a 3D printed case to create a wearable that was able to detect alcohol [3]. This was a proof-of-concept project that was able to show that commodity hardware was able to detect alcohol to a degree of accuracy. However, the device did not actually measure the consumption of the wearer instead it merely detects alcohol in the air.

We plan to build off these studies and more by creating a device that is able to detect alcohol consumption using commodity hardware and machine learning techniques. Though previous studies have used machine learning techniques to investigate accelerometer and gas sensors ability to detect alcohol consumption. We aim to replicate their results, and to investigate the ability of these techniques to detect consumption in real-time.

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